

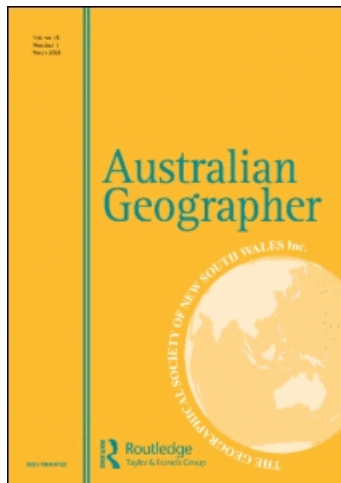
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Scenarios for Community-based Approaches to Biodiversity Conservation: a case study from the Wet Tropics, Queensland, Australia

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Scenarios for Community-based Approaches to Biodiversity Conservation: a case study from the Wet Tropics, Queensland, Australia

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ABSTRACT *Natural resource management approaches that deliver biodiversity conservation remain elusive, with evidence of a persistent implementation gap between biodiversity science and conservation projects. Scenarios have been identified as potentially useful in addressing the complex issues underlying this implementation gap, but have been infrequently applied to biodiversity conservation. Our paper reports on action co-research to develop, apply and assess the efficacy of scenarios within a community-based natural resource management (CBNRM) approach to biodiversity conservation at Mission Beach, a key site within the globally significant Wet Tropics bioregion. We focused on the capacity of scenarios to address the issues of contested interests and uncertainty, aiming specifically to engage the community to build a cohesive vision. The scenarios' headline messages included a projected substantial loss of habitat in coastal vegetation communities that are highly valued by all stakeholders. Our assessment identified that the use of scenarios fulfilled the intended aims, resulting in a vision for biodiversity conservation that has substantial community support. Three factors contributed to this efficacy of the scenarios: (1) the focus on threat; (2) biodiversity science integration; and (3) simplicity in presentation. Further investigation of the potential of scenarios as tools to overcome the implementation gap in biodiversity conservation is recommended.*

KEY WORDS *Regional planning; conservation planning; participatory scenario development; CBNRM; stakeholders; land use; Mission Beach.*

Introduction

Scenarios are gaining recognition as important tools for environmental planning that can address multiple challenges including community engagement, knowledge integration, conflict resolution, and decision making in the face of uncontrollable, irreducible uncertainty (Raskin *et al.* 2005). Scenarios provide insights into drivers of change, reveal the implication of current trajectories, and can illuminate options for action through presentation of different and competing visions of the future with vivid, clear and striking differences (Carpenter *et al.* 2006). Scenarios that link social, economic and environmental values and issues, based on both quantitative modelling and qualitative narratives, have been applied to consider global trajectories and international policy options since the 1970s, and are increasingly used at continental, regional and local scales (Patel *et al.* 2007; Raskin *et al.* 2005). In this paper we report on the use of scenarios in local-scale community-based natural resource management (CBNRM) at Mission Beach, a key site of contested land use and high biodiversity values in the Wet Tropics bioregion of north-east Queensland.

Community-based approaches to NRM have been mandated in Australia since 2002 through a program which devolved responsibilities from national and state governments to organisations with community-based governance structures, reflecting global trends to decentralised management across natural resource sectors (Robinson *et al.* 2009; Tacconi 2007). Community engagement methods that build capacity and integrate across institutions, strategies and knowledge systems are required to achieve social and environmental outcomes within CBNRM in Australia (Measham 2007; Morrison *et al.* 2004). CBNRM requires co-design for specific communities and natural resources using participatory tools that ensure integration of science, local and Indigenous knowledge, and recognise the diversity of human values associated with the environment (Berkes *et al.* 2007; Hill *et al.* 2010).

A rich body of literature exists on effective participatory tools for such integrated CBNRM, including visioning, scenarios, historical trends analysis, throw-away models, participatory mapping and problem co-analysis (Lynam *et al.* 2007; Sayer & Campbell 2004). However, these tools are still applied infrequently in biodiversity conservation projects, where the implementation gap remains problematic. More than two-thirds of the biodiversity conservation assessments in the peer-reviewed scientific literature have not resulted in any conservation action (Knight *et al.* 2008). Scenarios are of particular interest because of their identified potential to address issues such as uncertainty, contingency, reflexivity, and disparate conflictual interest groups, which typically plague biodiversity problems (Peterson *et al.* 2003). Scenarios have previously been applied and proven useful in the Wet Tropics for stimulating dialogue over sustainability in future landscapes, and building social capital for environmental planning broadly (Bohnet & Smith 2007). Our research sought to build on these outcomes through interrogating the identified potential of scenarios within a biodiversity conservation application at Mission Beach within the Wet Tropics bioregion (Hill *et al.* 2010).

The Wet Tropics bioregion is characterised by its tropical climate, spectacular scenery, cultural values, economic productivity and biodiversity whose global significance is recognised through protection of some 900 000 ha within the Wet Tropics World Heritage Area (WTWHA). Within the region, Mission Beach, the focus of this research, has been identified as a priority area for action on biodiversity

conservation, reflecting its significance for a range of biota, and in particular the endangered southern cassowary (*Casuarius casuarius johnsonii*) (FNQ NRM Ltd and Rainforest CRC 2004; Latch 2007). The goal of our action co-research was to develop, apply and assess the efficacy of scenarios within a CBNRM approach to biodiversity conservation. The scenarios as a tool focused on the identified biodiversity conservation problems of disparate conflictual interest groups and uncertainty, with two specific objectives: (1) to engage the community; and (2) assist in building a cohesive community vision (Peterson *et al.* 2003). Our paper presents insights in relation to threat, science integration and simplicity that emerged from our analysis, and concludes with a consideration of implications for future research.

Scenarios: history and relationship to CBNRM

The scenario method was first developed in response to problems in forecasting for challenging future issues including nuclear strategy and the impact of technology (Kahn & Wiener 1967; May 1996). Scenarios differ from forecasts by providing a plausible sequence of future events, rather than predictions, based on identification and incorporation of key drivers of change. Scenarios gained impetus in the 1970s for addressing growing global natural resource problems, characterised by uncertainty, multiple linkages to social, economic and cultural dimensions, including expanding human population and economies (Raskin *et al.* 2005). Scenario formulation techniques have developed in two main streams: deterministic, quantitative modelling, and qualitative descriptive analyses, with a more recent emphasis on participatory approaches in both (Rotmans *et al.* 2000). The recent Millennium Ecosystem Assessment (MEA) scenarios are at the forefront of integrated methods, using both rigorous modelling and rich, insightful narrative to address changes in ecosystem services and implications for human well-being (Carpenter *et al.* 2006).

The MEA was the first substantial exercise to address biodiversity issues through scenarios, and identified a high certainty of global species extinctions and loss of ecosystem services in all cases, with the highest rates of loss occurring in tropical forests and woodlands and warm mixed forest and savannas ecosystems (Peterson *et al.* 2003; Sala *et al.* 2005). This application triggered further interest in the development of both the technical biodiversity models (beyond species–area relationships) and the use of scenarios in participatory CBNRM approaches (Cumming 2007). The government-mandated arrangements for NRM in Australia rely on bringing disparate, adversarial interest groups and fragmented institutional arrangements within regional communities together to re-invigorate integrated action on urgent environmental problems (Lockwood *et al.* 2009; Robinson *et al.* 2009). A range of approaches are showing promise as tools for achieving this deliberative vision, including large-scale collaborative partnerships, web-based planning support tools (Nelson & Pettit 2004), community surveys (Measham 2007) and capacity-building measures (Robins 2008). Scenarios were identified as one of four key principles to link local communities into government NRM priorities in Indonesia (Wollenberg *et al.* 2009). Our study's goal of assessing the efficacy of scenarios as a tool within a CBNRM approach to biodiversity conservation builds on these and other recent initiatives that are applying scenarios in participatory NRM settings (Enfors *et al.* 2008; Newton *et al.* 2009).

Methods

Case study: history and geography

Biodiversity conservation activities at Mission Beach are embedded within a complex planning system aimed at sustainable management of natural and cultural resources in the Wet Tropics bioregion. The formulation and implementation of the *Sustaining the Wet Tropics: a regional plan for natural resource management 2004–2008* (Regional NRM Plan) achieved an unprecedented level of coordination between over 100 pre-existing plans, while the recent *Far North Queensland regional plan 2009–2031* (FNQ 2031 Plan) establishes clear patterns for future urban, rural and residential land uses (FNQ NRM Ltd and Rainforest CRC 2004; Minister for Infrastructure and Planning 2009). Mission Beach is recognised in both plans as a hotspot for action in biodiversity conservation, and is identified as a key site in the Cassowary Recovery Plan (Latch 2007). While the 60 per cent of Mission Beach habitat within the WTWHA is relatively well protected, the habitat outside the protected area estate has similar globally significant values, but is subject to ongoing loss, fragmentation and degradation, despite past efforts to reverse this decline. In 2006, a project was therefore initiated to create a new model and tools for a CBNRM approach to biodiversity conservation at Mission Beach (Weston 2006). The project was championed by Terrain NRM, the community-based organisation mandated by the Australian government to deliver regional NRM (Robins & Dovers 2007).

The Mission Beach study area of some 24 000 ha is located approximately 120 km south of Cairns, and is bounded by Maria Creek in the north, the Hull River to the south, the Walter Hill Range to the west and the Coral Sea to the east (see Figure 1). Mission Beach now falls within the newly reformed Cassowary Coast Regional Council. It previously lay across the boundaries of the now amalgamated Johnstone and Cardwell Shires. Mission Beach consists of four smaller villages—Bingil Bay, North Mission Beach, South Mission Beach and Wongaling Beach—interspersed with coastal rainforest communities of conservation significance. Four topographic units dominate: lowland plains, coastal ranges, tidal flats and flood plains. Land uses within the area are primarily agricultural, conservation, residential and tourist.

Mission Beach, in common with much of coastal Australia, is currently experiencing development pressure associated with sea-change and tree-change human population movements (Gurran & Blakely 2007). Coastal regions have experienced strong human population growth in recent decades, in part due to their aesthetic and environmental attributes, leading to an increase in coastal dwelling construction and economic activity. Demand for residential and tourism development is driven by institutional investors, second-home buyers, and retirees cashing in on high city property values, increasing the expense of low-cost housing (Gurran & Blakely 2007). At Mission Beach, this transformation process is threatening important biodiversity values—in an area less than 30 000 ha, our study site includes 36 per cent of Australia's bird species, 13 per cent of remaining wet tropical lowland rainforest, and 5 per cent of all Australian vascular plant species. It is a significant site for conservation of the endangered southern cassowary, which is threatened by habitat loss, fragmentation and degradation, roads and traffic, dog attacks, hand feeding, diseases and natural catastrophes, particularly cyclones (Latch 2007). Cassowary habitat in the Wet Tropics has been cleared extensively, with up to two-thirds of all lowland habitat

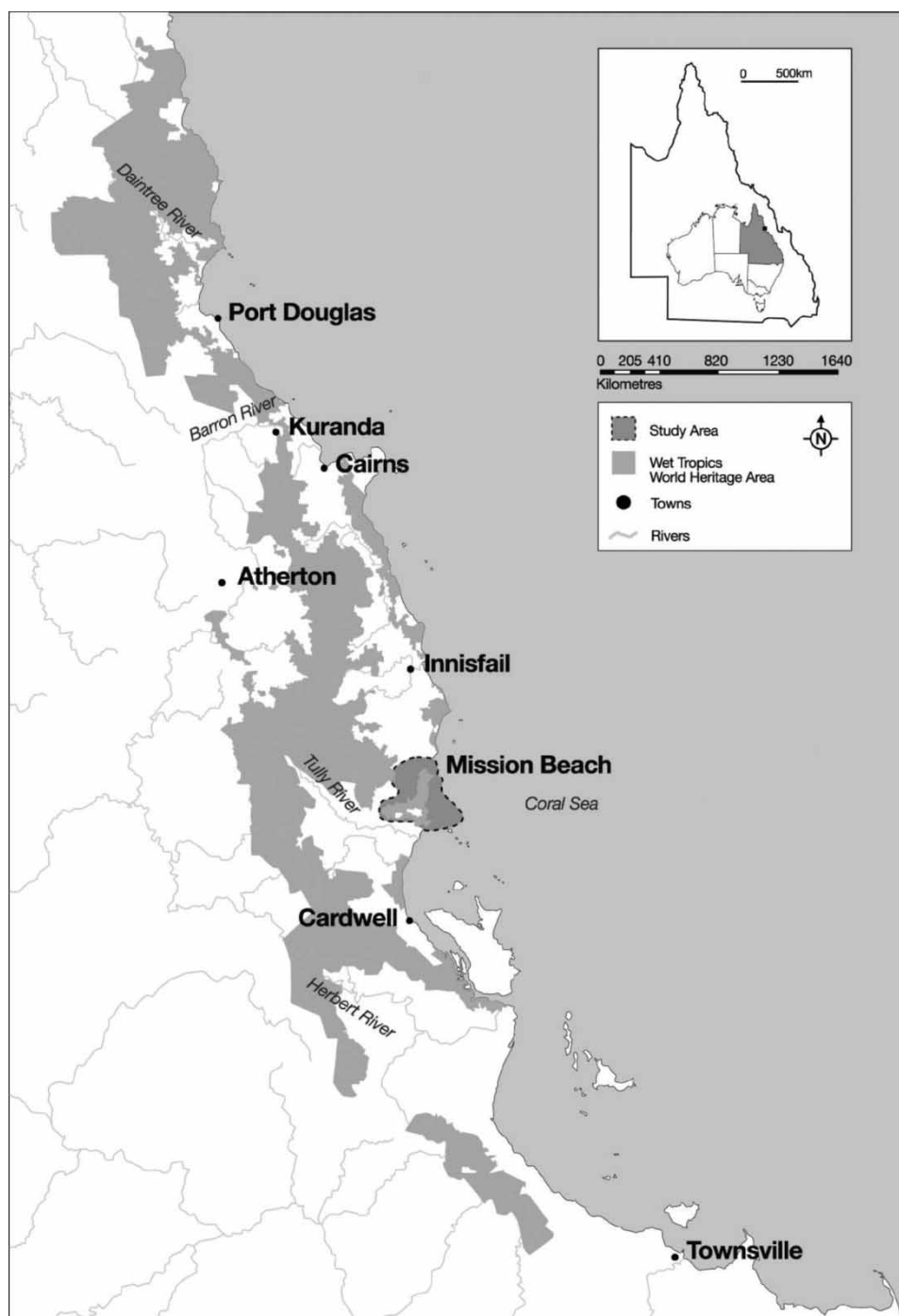


FIGURE 1. Location of Mission Beach study area, within Wet Tropics World Heritage Area.

cleared across the region and more than 30 per cent in the Mission Beach study area (Williams *et al.* 2008). Approved and pending development applications in the Mission Beach study area continue to threaten cassowary habitat.

Action co-research

An action co-research approach was used for the study, combining real-world experimentation with bringing researchers and practitioners together in an equitable working relationship for co-generation of new knowledge (Greenwood & Levin 2007; Wilmsen *et al.* 2008). The term 'co-research' signals our attention to situate our approach as collaborative and empowering, responding to critiques of the diversity within the participatory action research field, and the opportunities to build discursive community practice for NRM (Kesby *et al.* 2007; Nelson & Pettit 2004). Our study of scenarios was embedded within a larger research project aimed at creating a place-based learning community and dialogic network to co-produce knowledge (Hill *et al.* 2010). Our core co-research team included three people from Terrain NRM and three scientists. The Mission Beach Habitat Network Action Committee (hereafter committee) provided the forum for a wider dialogue with civil society, industry, scientific and government actors who had decision-making roles at local, regional and national scales.

Our initial scenario development approach created sets of scenarios that were vivid and different, in keeping with the guidance in the scientific literature (Carpenter *et al.* 2006; Peterson *et al.* 2003; Rotmans *et al.* 2000). We considered a wide range of potential responses to projected possible future events by societies, economies, and ecosystems, aiming for rigour, consistency, transparency and imagination, while moving beyond 'business-as-usual' formulations. Data sources for scenario formulation and assessment of efficacy included biophysical and social data held by the core co-research team, policies, plans, media releases, annual reports, terms of reference, minutes of meetings, consultation reports, historical essays, and data held by researchers and others represented on the committee. Participant observation was undertaken by a member of the core co-research team at the community workshop attended by approximately 50 people and in numerous smaller meetings with stakeholders (Evans 1988). The assessment of efficacy of the scenarios used a simple two-stage objective-oriented evaluation approach: (1) determining whether the objectives as originally set out were achieved; and (2) undertaking content analysis and theme identification of the documentary and participatory observation data to interrogate the factors behind the performance of the scenarios (Kumar 2005). Validity of the content analysis and theme identification was established through convergent triangulation between multiple data sources and the co-research team's regular conduct of critically reflexive reviews (Robinson 1998).

Results

Phase one: preparing for the participatory scenario development

The scenario development process took place over an 18-month period (see Figure 2), beginning with an institutional analysis. Key factors identified as hampering previous biodiversity conservation efforts included: divergent goals of local, state and federal agencies; insufficient resources for responsible organisations and authorities; diverse aspirations of community groups, individuals, industries and others; gaps in current institutional arrangements; and a lack of coordination and collaboration between institutions. Stakeholders' views ranged across a spectrum: (1) facilitating development while relying on existing national parks to protect

Scenario Development Time

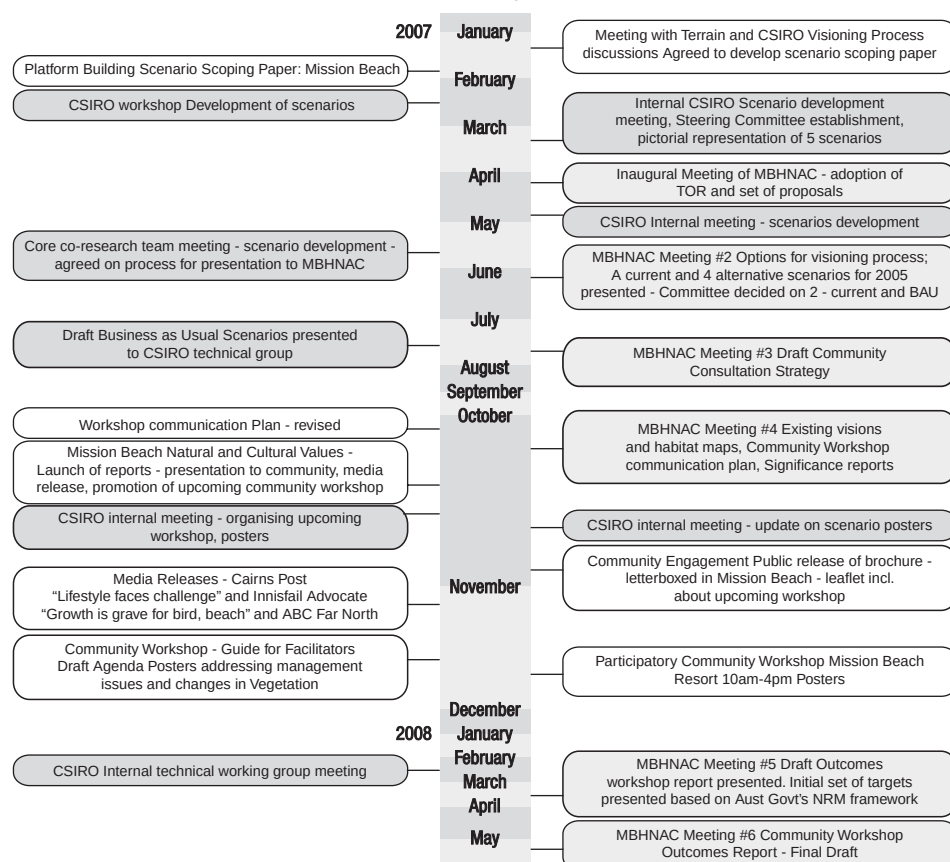


FIGURE 2. Scenario development timeline (MBHNAC refers to Mission Beach Habitat Network Action Committee).

habitat; (2) increasing regulatory protection for habitat within a clear threshold to limit development; and (3) imposing an immediate moratorium on any further development that impacts habitat. The Queensland government's post-cyclone cassowary feeding and relocation program highlighted the controversies, with considerable media debate about the choice between ensuring cassowary survival through native habitat protection or through active feeding and management. A number of instruments were proposed to address these divisions: a local action committee, common community vision, values significance assessment, broad community engagement, institutional brokering and science partnerships (Hill *et al.* 2009).

Development of a common vision between key stakeholders in NRM is recognised as a useful means of overcoming conflicting community perceptions, diverse aspirations and divergent goals (Moore *et al.* 1999). We considered a number of different approaches to engaging stakeholders in the formulation of such a vision: community-wide visioning, charette, future search conference, strategic perspectives analysis, scenarios and first-cut draft vision workshop (Moore *et al.* 1999; Susskind *et al.* 1999). Scenarios were viewed as potentially highly salient to the Mission Beach context because of their power in stimulating people to evaluate and reassess their

beliefs about the system, thereby encouraging reframing and movement from conflict to cohesion (Peterson *et al.* 2003; Rydin 2003). Early scoping of scenarios in Mission Beach identified their potential for also integrating biodiversity science associated with thresholds in traffic and other threatening processes.

Phase two: the building of the scenarios

The building of the scenarios followed the first five stages of the process outlined by Peterson *et al.* (2003):

- (1) identification of the focal issues;
- (2) assessment of the current system, key drivers of change/uncertainties, measures of effectiveness, potential thresholds for regime shifts;
- (3) identification of possible alternatives;
- (4) building of spatially explicit scenarios; and
- (5) analysis of scenarios and recommended policy choices.

The local action committee was first brought together in May 2007 to oversee and guide the planning process, including the scenarios. Our stakeholder analysis (Reed 2008) identified key actors from a range of sectors resulting in a committee including representatives from three industry, two local government, four state government, one national government, two research and five civil society groups, including one Traditional Owner organisation. The committee's role was to facilitate community ownership of the process with members undertaking to liaise with their organisations and networks.

At its first meeting, the committee adopted the overall goal 'to develop a network of natural habitat that is ecologically viable and protects identified community values' and endorsed a proposal from the core co-research team to co-develop a community-based vision, and requested the team propose a relevant process. A proposal to use scenarios to develop the community vision was formulated, using the committee's overall goal as the focal issue. Assessment of the current system identified key drivers of change, with associated measures of relevance to evaluating the impacts of different scenarios (see Table 1). Cassowary conservation was recognised as a key indicator and relevant planning focus for consideration of natural and cultural values, and a collaborative focal species approach later emerged within the larger project in response to this recognition (Hill *et al.* 2010). Four scenarios were proposed, each reflecting alternative futures that gave differential weighting to biodiversity, agriculture, and residential development options (Table 2).

However, the committee viewed this approach of four scenarios as overly complex and not focusing sufficiently on the key issue of habitat protection. Instead, the committee agreed that the purpose of the scenarios should be to support a comprehensive community engagement strategy that would: (1) provide opportunities for people to say what they want and influence decisions; (2) ensure the wider community is informed about the planning exercise; and (3) develop a strong community vision that would attract wider interest and support. Participatory development of the scenarios with the community was rejected as time-wasting. Instead, the committee asked the team to develop a revised proposal for the formulation of two scenarios, one depicting the current situation, and one showing business as usual (BAU) in 2025.

TABLE 1. Primary key drivers of change, measures of these, methods and justification

Key driver	Measure (final current and BAU scenarios)	Methods/datasets
Population growth and infrastructure	• Residential population • Tourist population • Tourist establishments	• <i>Current</i> 2006 ABS data; <i>BAU</i> modelled from adjusted 2001–06 change rates (Williams <i>et al.</i> 2008) • As above • Mission Beach Tourism (industry data)
Human settlement density	Intensive and other land uses	<i>Current</i> QLUMP 2004 data; <i>BAU</i> modelled from 1999–2004 trends
Key industries	Key employment	<i>Current</i> 2006 ABS data; <i>BAU</i> modelled from adjusted 2001–06 change rates
Traffic and road management	• Vehicles per day • Cassowary underpasses; roadkill hotspots	<i>Current</i> Department of Main Roads traffic data; EPA/QPWS data 1992–2006 on cassowary road deaths at MB; <i>BAU</i> modelled from these
Dogs and other invasive species	• Registered dogs • Qualitative statement on invasive species	<i>Current</i> MB population-equivalent proportion of total dog registrations; <i>BAU</i> modelled on 3% growth rate. Qualitative based on development trends
Human interactions with cassowaries	Qualitative	Descriptions based on records of current behaviours (Latch 2007)
Conservation regulation and funding support	Qualitative and spatially explicit quantitative	Qualitative based on current situation and trends in restoration and recovery plans and Djiru people's aspirations. Table quantifies the areas protected by a variety of regulatory/legal tenure (EPA 2006)
Natural disasters (e.g. cyclones)	Qualitative	Qualitative observations on effects of Cyclone Larry included
Conservation effort	Spatially explicit quantitative	Effect of conservation effort on the biodiversity status of regional ecosystems is presented in a table

TABLE 2. Original scenarios proposed to Mission Beach Habitat Network Action Committee (Hill *et al.* 2007)

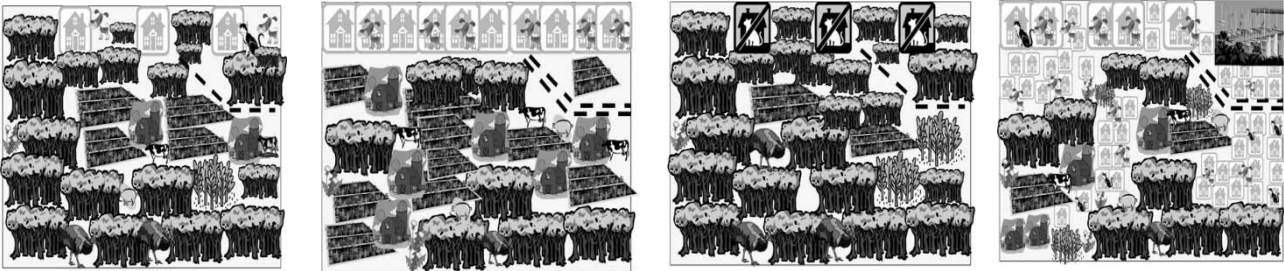
				
Key drivers	Current situation	Baseline scenario	Backyards and biodiversity	Noosa in the North
Population growth	Very high population growth	Very high population growth of 3% continues	Growth levels slowed to 1%	Population tripled or more by 2025
Human settlement density	Low-rise, unit development, strip development emerging	Medium-rise, unit development Strip development between villages	Little changed	High density occupancy over all available cleared space
Key industries	Tourism, farming	Tensions between mass and nature-based tourism; farming declining	Stagnation of both tourism and farming	Economy booming as mass tourism replaces nature-based tourism
Traffic and road management	One scenic road and one two-lane road, both with cassowary calming	A two-lane road and four-lane highway with little or no traffic calming	Same roads but with network of cassowary safety measures	One scenic road and one two-lane road, both with network of cassowary safety measures

TABLE 2 (Continued)

Key drivers	Current situation	Baseline scenario	Backyards and biodiversity	Noosa in the North
Dogs and other invasive species	Little regulation, some limits on dog numbers and licences	Little changed	Impacts of pets and invasive species minimised through by-laws	No controls on dogs, cats, weeds, etc.
Human interactions with cassowaries	Frequent feeding of cassowaries	Frequent feeding of cassowaries	Heavy fines prevent cassowary feeding	Frequent feeding of cassowaries
Government regulation and funding	Fragmented with limited effectiveness	Fragmented with limited effectiveness	Comprehensive and effective with well-funded restoration efforts	Government regulation remains in place but focus shifts to funding through the market
Market-based instruments	Offsets started with some uptake	Offsets now being widely used in rampant development	Some offsets but conservation still reliant on government funds	Offsets now being widely used in rampant development, funded through carbon trading
Natural disasters	Cyclone Larry devastated Mission Beach in 2006	A second category five cyclone 15 years after Larry	A second category five cyclone 15 years after Larry	A second category five cyclone 15 years after Larry
Conservation effort	Substantial regulatory but poor incentive-based conservation	Weak effort overall	Very strong regulatory but little incentive-based conservation	Strong restoration funded through the market
Community cohesion and conservation	Poor cohesion	Increased cynicism and division	Conflict about over-regulation and economic stagnation	Conflict over development impacts and loss of social fabric

The revised proposal for two scenarios was presented in August 2007, and endorsed by the committee for use in the workshop. The meeting reached consensus around a simple poster format, including a qualitative statement, a map, relevant photographs, and sets of quantitative data. The two scenarios finally developed and used in the workshop were:

- (1) a scenario reflecting current conditions (current scenario); and
- (2) a scenario modelling conditions in 2025 if current trends in land use and population continue without any specific policy interventions (2025 BAU scenario).

Table 3 shows how the drivers were measured, together with justification for the measurement approach, for these two scenarios. The original set of drivers was narrowed to focus on issues most salient to a BAU approach. The committee's requirement for simplicity, clarity and a poster format underpinned the design of the scenario presentation (see Figure 3). Six sets of information were incorporated into an integrated poster layout:

- (1) a qualitative statement addressing the key drivers identified through the scoping study and exploratory phases of the planning exercise;
- (2) a set of quantitative data addressing these key drivers;
- (3) spatial land-use data in five categories;
- (4) analysis of conservation status in terms of (primarily) tenure-based institutional arrangements for protection and restoration;
- (5) analysis of regional ecosystem biodiversity status; and
- (6) photographic images to illustrate key messages.

Table 3 presents the qualitative and quantitative information captured within each scenario, and the spatial pattern of population and intensive land use are shown in Figures 4 and 5. The modelling of a continuation of current trends in population growth and land use change by 2025 identified a number of headline results which included:

- Mission Beach will have around 3000 more people (a doubling of the current population, highlighted in Figure 5, primarily along the coast);
- an additional 528 more hectares of intensive land use will bring additional dogs, cats and traffic pressures on habitat and clearing of 397 ha of intact remnant vegetation;
- loss of habitat through clearing will be concentrated on highly valued coastal vegetation communities, including some 190 ha of already threatened regional ecosystems;
- Mission Beach in 2025 is likely to have more restored and protected habitat areas for its unique plants and wildlife; and
- the tourism industry will be the region's largest employer, with the traditional industries of agriculture and fisheries declining.

The extent to which these changes will impact on biodiversity values was assessed through the analysis of the effect on ecosystem endangerment. Regional ecosystem mapping classifies vegetation communities as endangered, of concern, and not of concern (EPA 2006). The projected land-use requirement of residential and visitor populations to 2025 showed a loss of 19 ha of 'endangered' and 190 ha of 'of-concern' regional ecosystems, some associated with road developments, but most

TABLE 3. Current and business-as-usual (BAU) scenarios

Drivers	Current		BAU 2025	
	Quantitative	Key qualitative	Quantitative	Key qualitative
Population growth and infrastructure	Residents—4103 Tourists—1646 Tourist establish—73	The relaxed lifestyle is under threat from increasing development pressures from tourism and residential units	6582 2541 129	Very high population growth (over 3%) continues within the faster growing parts of Mission Beach
Human settlement density	Intensive—622 ha	Vying for space in the narrow coastal belt of habitat	Intensive—ha	Coastal strip dense sea-change residential location
Key industries	17% agriculture, forestry and fishing 12% retail 11% accommodation, restaurants & cafés	Trends of declining and increasing jobs in employment in agriculture tourism and construction	5% agriculture, forestry and fishing 12% retail 25% accommodation, restaurants & cafés	Tourism largest employer, construction strong, agricultural and fisheries weak
Traffic and roads	3600 vehicles per day	Traffic-calming devices and a faun underpass are being trialled	Additional 5789 cars per day	Higher levels of road traffic
Dogs and invasive species	1957 registered dogs	Dogs, pigs and weeds perceived as a threat	3400 registered dogs	Increasing numbers of dogs, pigs and weeds
Government regulation	34% National Park 2% Reserves	A significant area is protected by legislated tenure	N/A	A large portion of coast vegetation cleared

TABLE 3 (*Continued*)

Drivers	Current		BAU 2025	
	Quantitative	Key qualitative	Quantitative	Key qualitative
	0.7% Covenants/Nature Refuges	conservation status		with more effective enforcement of laws protecting some remnant on the coast
Natural disasters	Cyclone Larry—2006	Cyclone-related impacts including the death of 15 adult cassowaries in an 18-month period	N/A	Increased frequency of future natural disasters under climate change
Conservation effort	Regional Ecosystems outside conservation areas—5618 ha	Restoration plans being formulated to connect isolated fragments of remnant vegetation and improve habitat connectivity	Regional Ecosystems outside conservation areas—4712 ha	Green corridors through the rural matrix. Fragmented coastal matrix of wetlands and mixed rainforest and open forest habitats

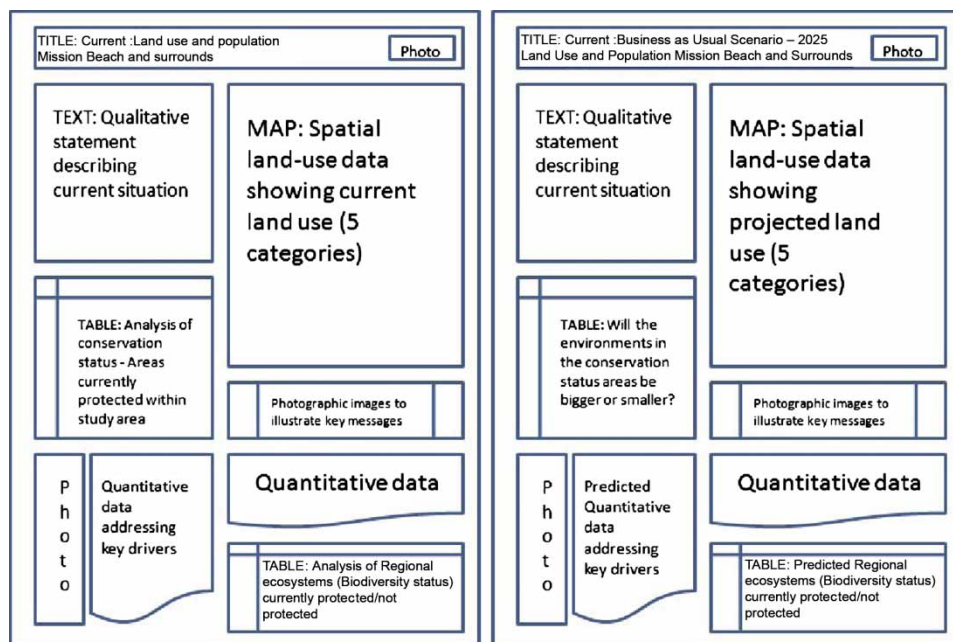


FIGURE 3. Poster mock-ups representing the two scenarios and format.

attributed to new urban or rural residential developments along the coast. Furthermore, an estimated 858 ha of disturbed vegetation could be cleared through intensification of agricultural practices by 2025 (Williams *et al.* 2008). There was potential for revegetation of existing clearings in conservation areas (51 ha), but this was exceeded by new clearing for new roads or widening/realignment of existing roads (116 ha).

Phase three: applying the scenarios in a community workshop process

The primary purpose of the community workshop where the scenarios were applied was to initiate the process of developing an agreed vision. Selection of workshop participants aimed to engage key actors from sectors important to biodiversity conservation. The committee drove this selection process, with all members providing lists of possible attendee names, which were circulated for comments and additions, and a final list approved by consensus. Participation in the workshop was by invitation only with no general public invitation. A post-workshop review by the committee identified that while the selection process had resulted in a good mix across civil society, industry, government, Traditional Owner and research communities, there was under-representation of youth and women, which in part reflects the demographic profile of Mission Beach residents.

The scenarios were released publicly 2 weeks prior to the workshop, and received substantial interest from the local media, including a feature in the major regional newspaper. The workshop was carefully structured in two phases. In the first phase, the scenarios were presented along with information about the outcomes of the significance assessment of the identified community values associated with habitat. The workshop was then broken into seven small groups to address the following focus question:

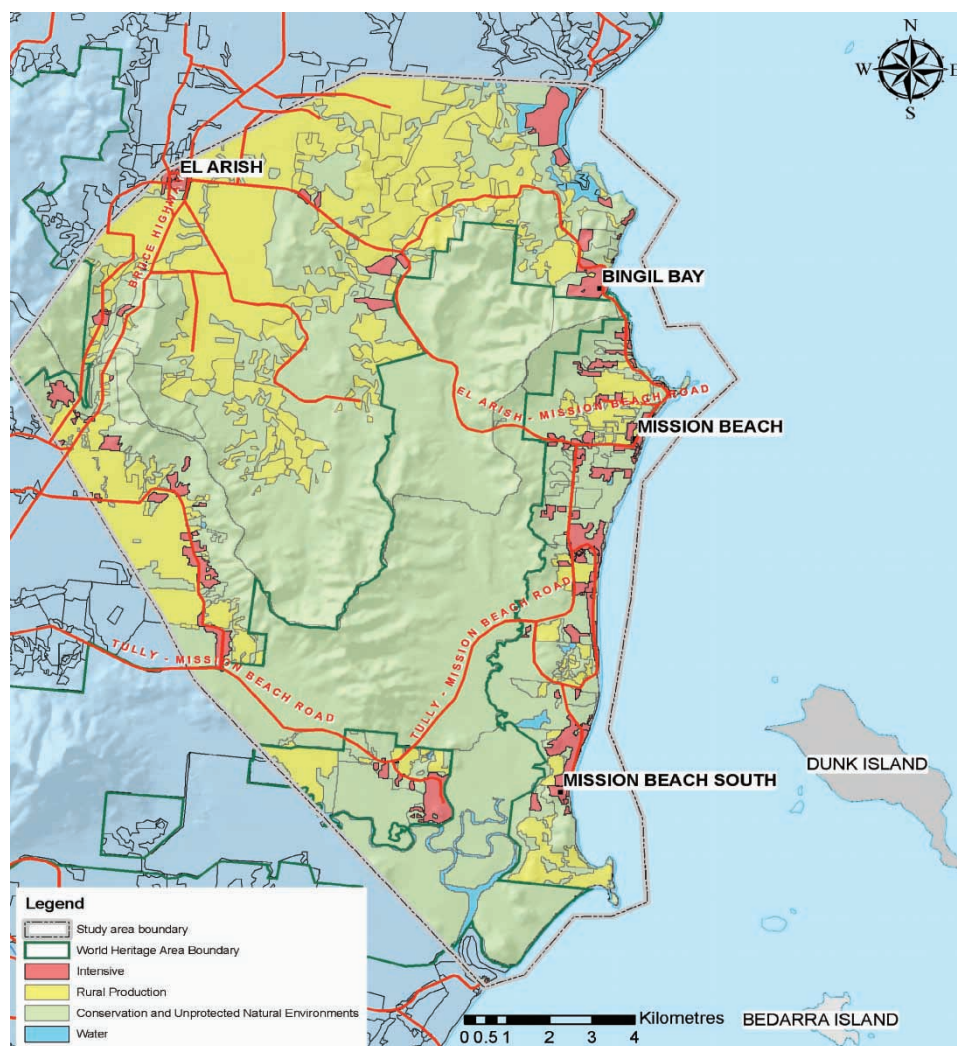


FIGURE 4. Current land use.

Our goal is to develop a habitat network that is ecologically viable and protects the identified community values. Imagine that by 2025 we have achieved our goal. What does Mission Beach look like then? How does it differ from the BAU 2025 scenario?

In the second phase of the workshop, further information was presented regarding management issues highlighted by the BAU scenario, followed by small break-out groups to address the question:

Again imagine that by 2025 we have achieved our goal of an ecologically viable habitat network that protects the identified community values. What are the prime issues we need to address? What steps did we take to address these issues?

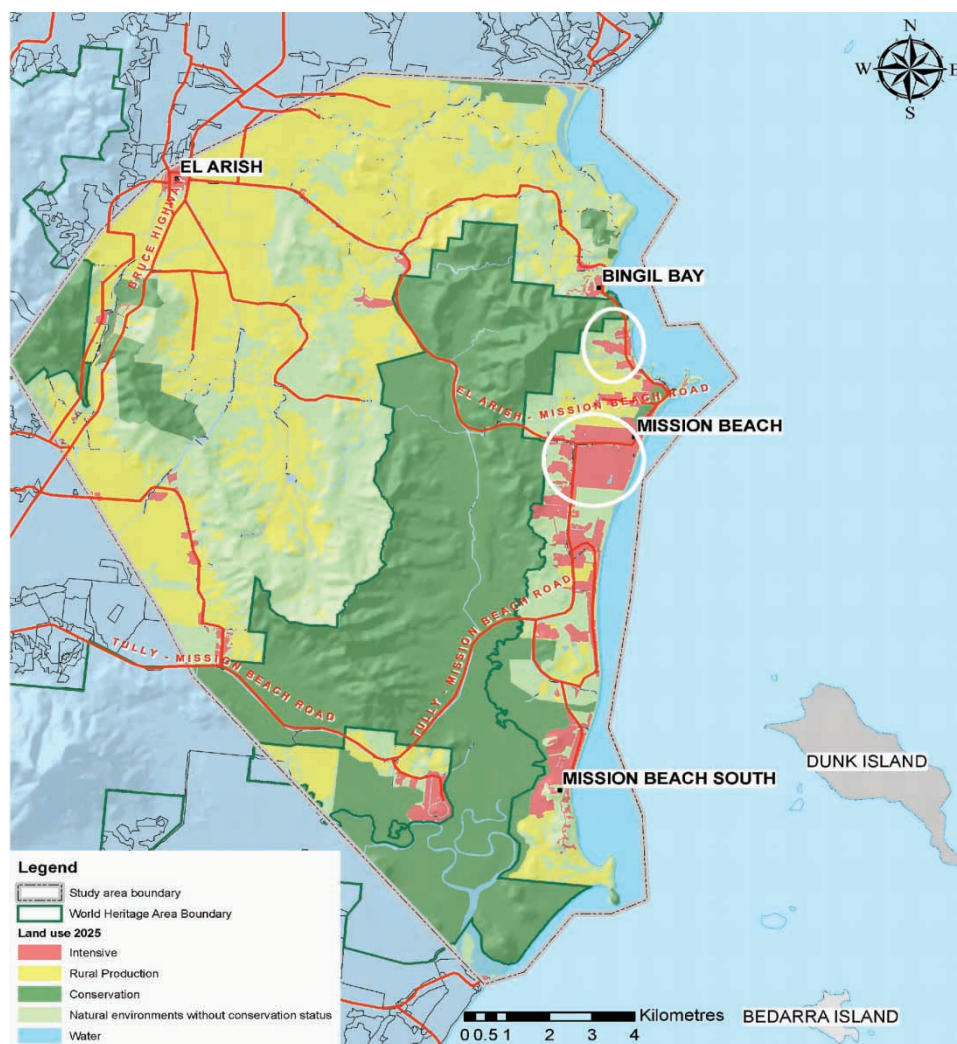


FIGURE 5. Business-as-usual scenario—2025 land use.

Several members of the committee acted as facilitators at the workshop, guided by a handbook co-produced between the researchers and the committee.

Interrogation of the scenarios elicited a strong reaction from the workshop participants, identifying an overall vision for the future as one in which the BAU scenario does not eventuate. Comments included:

Nothing we want is connected with the BAU—we want conservation and sustainability with more cassowaries, more habitat corridors, and more production from new markets.

BAU can not be realistic—we want a clean green image, with a clearly defined tourism market, common values around sustainability.

Key themes that emerged in relation to the desired future vision included concepts of sanctuary, of an environmentally focused way of life, and a strong community achieving sustainability. Detailed steps to achieve the vision were captured in relation to the management of traffic, tourism, residential and infrastructure development, agriculture, management by Traditional Owners, habitat protection and restoration actions, exotic species, institution building and ensuring economic viability. A report on the workshop outcomes was finalised together with the committee, and posted on the Terrain Ltd website in March 2008 (Hill & O'Malley 2008). This report synthesised an agreed community vision from the major themes at the workshop which was subsequently endorsed by the committee:

Mission Beach is a sanctuary for wildlife and habitat; its defining feature is a strong human community that acts to protect its special values. Mission Beach is an exemplar of sustainable living, both environmentally friendly and culturally diverse. Mission Beach has a tropical landscape character where urban, farming, and forest communities blend to maintain a harmonious setting with strong visual appeal.

Committee members undertook a review of the community workshop in March 2008. Both process and outcomes were identified as positive, based on feedback to the committee members from their networks, including both workshop participants and the wider community.

Discussion

Efficacy of the scenarios

The objective-oriented evaluation identified substantial evidence that the objectives as originally set out had been achieved. The community engagement was demonstrated by attendance and participation at the workshop, and positive feedback from stakeholder networks identified through the committee review process. A community vision was developed and endorsed by all the disparate interest groups on the committee and present at the workshop. However, the scenarios formed one relatively small part of a broader engagement strategy using numerous other tools including media releases, direct mail, web-based interaction, public launches and one-to-one meetings. The second part of the assessment therefore focused on analysing factors underlying the contribution of the scenarios to the achievement of these objectives. This analysis identified three factors of importance: (1) the focus on threat; (2) biodiversity science integration; and (3) simplicity in presentation.

Threat

The committee's original direction to focus on the current and BAU scenarios was based on a perception that BAU would provide a justification for the disparate interest groups to start working together. This proved to be true, with a common strong negative reaction to the outcomes of BAU among all the interest groups present at the community workshop. Use of BAU in this context has much in common with worst-case scenario planning, a popular approach to military planning (Edwards 1999). Worst-case scenario planning is considered appropriate when threats are urgent, i.e. climate change impacts and security implications (Edwards

1999). The underlying rationale is that an optimistic view of military threat could lead a nation to disaster. Threats are also recognised as a means of mobilising community cohesion. Communities are formed and shaped by opposition, engaging powerful rhetorics of place and 'home' which inspire cohesion in resistance (Dalby & Mackenzie 1997). The efficacy of BAU in highlighting threats to all interest groups was a key factor in its ability to promote the formulation of a cohesive community vision. Given the high threats to biodiversity identified in all global Millennium Ecosystem Assessment scenarios and the evidence for an accelerating humanity-driven extinction crisis, further investigation of the application of worst-case scenario planning in the field of biodiversity conservation is warranted (Sala *et al.* 2005).

Biodiversity science integration

The action co-research approach resulted in the co-design of the scenarios between the committee and the scientists. While the practitioners' involvement ensured the focus on threat, thereby enabling the scenarios' contribution to building cohesion, the science contribution built confidence in the headline results from the scenarios. These headline results were widely reported in the media, stimulating considerable interest and providing the foundation for workshop participants to commonly perceive that the threat being faced was real. This common perception of threat reflects lowered community uncertainty about likely impacts of development on biodiversity.

Simplicity

The need to reduce complexity and present simple, transparent information was a consistent theme of the committee in relation to the scenarios, and indeed to all aspects of planning. This requirement for simplicity from practitioners intersects with the research community's goals to integrate increasingly complex quantitative biodiversity and other models with rich qualitative narrative in scenarios (Cumming 2007; Newton *et al.* 2009). The key characteristics for scenarios sought by the practitioners in our case study—limited diversity, extrapolations of current trends, unimaginative—are viewed as deficiencies in the scientific literature (Peterson *et al.* 2003). The importance of simple visual images and the usefulness of simple forecasting processes in *vision making* have been previously identified in scenario research (Bohnet & Smith 2007; Patel *et al.* 2007). Social context clearly should be an important determinant of the balance between rigorous quantitative modelling and qualitative descriptive presentations in scenarios—approaches like the MEA that integrate the two will not always be necessary or beneficial.

Conclusion

Our application of scenarios as a tool within a CBNRM approach to biodiversity conservation confirmed their relevance and ability to address the uncertainty and contested interests that plague biodiversity problems. The scenarios performed well as a tool for galvanising community engagement and building a cohesive community vision. Two scenarios were developed which demonstrated to the community the implications of the BAU trend continuing, including substantial impacts on biodiversity. This threat provoked a common and strongly negative reaction at the community workshop, bringing people together around a cohesive

community vision. Science integration within the development of the scenarios built confidence that the threats faced were real.

A need to reduce the number and complexity of scenarios was identified early in the study through the action co-research approach, which resulted in the refinement of scenarios and the production of two simple scenarios. This required simplification highlights a gap between current guidelines about scenarios in the scientific literature and the practical focus of practitioners. Interrogation of the social context for scenario application should occur before identifying the balance of rigorous quantitative modelling and qualitative descriptive presentations in scenario formulation.

The scenario technique combined with the participation of stakeholders proved to be a useful tool for reducing the contested interests which pose a barrier to implementation of biodiversity conservation. It enabled the community to envision a desired common 2025 future. Scenario building and visioning processes which engage community and integrate science can support effective consensus building in CBNRM approaches to biodiversity conservation, which in turn helps to overcome the implementation gap between conservation assessment and real-world conservation action. Further investigation of scenarios, including the use of threat-focused and worst-case approaches, is recommended as a means of building more effective tools for biodiversity conservation.

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REFERENCES

- BERKES, F., ARMITAGE, D.A. & DOUBLEDAY, N.C. (2007) 'Synthesis: adapting, innovating, evolving', in Armitage, D.A., Berkes, F. & Doubleday, N.C. (eds) *Adaptive co-management collaboration, learning and multi-level governance*, UBC Press, Vancouver, pp. 308–27.
- BOHNET, I. & SMITH, D.M. (2007) 'Planning future landscapes in the Wet Tropics of Australia: a socio-ecological framework', *Landscape and Urban Planning* 80, pp. 137–52.
- CARPENTER, S.R., BENNETT, E.M. & PETERSON, G.D. (2006) 'Scenarios for ecosystem services: an overview', *Ecology and Society* 11, p. 29.
- CUMMING, G. (2007) 'Global biodiversity scenarios and landscape ecology', *Landscape Ecology* 22, pp. 671–85.
- DALBY, S. & MACKENZIE, F. (1997) 'Reconceptualising local community: environment, identity and threat', *Area* 29, pp. 99–108.
- EDWARDS, M.J. (1999) 'Security implications of a worst case scenario of climate change in the south-west Pacific', *Australian Geographer* 30, pp. 311–30.

- ENFORS, E.I., GORDON, L.J., PETERSON, G.D. & BOSSIO, D. (2008) 'Making investments in dryland development work: participatory scenario planning in the Makanya catchment, Tanzania', *Ecology and Society* 13, p. 42.
- EPA (2006) 'Protected areas of Queensland—estates, version 12.4.1—31st December 2006', *Digital vector data*.
- EVANS, M. (1988) 'Participant observation', in Eyles, J. & Smith, D.M. (eds) *Qualitative methods in human geography*, Polity Press, Cambridge, pp. 197–218.
- FNQ NRM LTD AND RAINFOREST CRC (2004) *Sustaining the Wet Tropics: a regional plan for natural resource management 2004–2008*, FNQ NRM Ltd, Innisfail.
- GREENWOOD, D.J. & LEVIN, M. (2007) *Introduction to action research*, Sage, Thousand Oaks, CA.
- GURRAN, N. & BLAKELY, E. (2007) 'Suffer a sea change? Contrasting perspectives towards urban policy and migration in coastal Australia', *Australian Geographer* 38, pp. 113–31.
- HILL, R., BOHNET, I. & BUTLER, J. (2007) *Scenario scoping paper Mission Beach*, report to the Marine and Tropical Science Research Facility.
- HILL, R. & O'MALLEY, T. (2008) *Mission Beach habitat network action plan community workshop outcomes final draft*, CSIRO Sustainable Ecosystems and Terrain NRM, Cairns.
- HILL, R., WILLIAMS, K.J., PERT, P.L., GRACE, R., O'MALLEY, T. & JENKINS, S. (2009) *Mission Beach habitat network action plan public consultation draft*, CSIRO and Terrain NRM, Cairns.
- HILL, R., WILLIAMS, K.J., PERT, P.L., ROBINSON, C.J., DALE, A.P., WESTCOTT, D.A., GRACE, R.A. & O'MALLEY, T. (2010) 'Adaptive community-based biodiversity conservation in Australia's tropical rainforest', *Environmental Conservation* 37, pp. 73–82.
- KAHN, H. & WIENER, A.J. (1967) *The year 2000: a framework for speculation on the next thirty years*, Macmillan, New York.
- KESBY, M., KINDON, S. & PAIN, R. (2007) 'Participation as a form of power: retheorising empowerment and spatialising participatory action research', in Kindon, S., Pain, R. & Kesby, M. (eds) *Participatory action research approaches and methods connecting people, participation and place*, Routledge, Oxford, pp. 19–25.
- KNIGHT, A.T., COWLING, R.M., ROUGET, M., BALMFORD, A., LOMBARD, A.T. & CAMPBELL, B.M. (2008) 'Knowing but not doing: selecting priority conservation areas and the research-implementation gap', *Conservation Biology* 22, pp. 610–17.
- KUMAR, R. (2005) *Research methodology*, Pearson Education Australia, Frenchs Forest.
- LATCH, P. (2007) *National recovery plan for the southern cassowary Casuarus casuarus johnsonii 2007–2011*, report to Department of the Environment, Water, Heritage and the Arts, Canberra, Environmental Protection Agency, Brisbane.
- LOCKWOOD, M., DAVIDSON, J., CURTIS, A., STRATFORD, E. & GRIFFITH, R. (2009) 'Multi-level environmental governance: lessons from Australian natural resource management', *Australian Geographer* 40, pp. 169–86.
- LYNAM, T., DE JONG, W., SHEIL, D., KUSUMANTO, T. & EVANS, K. (2007) 'A review of tools for incorporating community knowledge, preferences, and values into decision making in natural resources management', *Ecology and Society* 12, p. 5.
- MAY, G.H. (1996) *The future is ours: forecasting, managing and creating the future*, Praeger, Westport, CT.
- MEASHAM, T.G. (2007) 'Building capacity for environmental management: local knowledge and rehabilitation on the Gippsland Red Gum Plains', *Australian Geographer* 38, pp. 145–59.
- MINISTER FOR INFRASTRUCTURE AND PLANNING (2009) *Far North Queensland regional plan 2009–2031*, Queensland Government Department of Infrastructure and Planning, Brisbane.
- MOORE, C.M., LONGO, G. & PALMER, P. (1999) 'Visioning', in Susskind, L., McKernan, S. & Thomas-Larmer, J. (eds) *The consensus building handbook*, Sage, Thousand Oaks, CA, pp. 557–91.
- MORRISON, T.H., McDONALD, G.T. & LANE, M.B. (2004) 'Integrating natural resource management for better environmental outcomes', *Australian Geographer* 35, pp. 243–58.
- NELSON, A. & PETTIT, C. (2004) 'Effective community engagement for sustainability: wombat community forest management case study', *Australian Geographer* 35, pp. 301–15.

- NEWTON, A.C., CAYUELA, L., ECHEVERRÍA, C., ARMESTO, J.J., CASTILLO, R.F.D., GOLICHER, D., GENELITTI, D., GONZALEZ-ESPINOSA, M., HUTH, A., LÓPEZ-BARRERA, F., MALIZIA, L., MANSON, R., PREMOLI, A., RAMÍREZ-MARCIAL, N., BENAYAS, J.M.R., RÜGER, N., SMITH-RAMÍREZ, C. & WILLIAMS-LINERA, G. (2009) 'Toward integrated analysis of human impacts on forest biodiversity: lessons from Latin America', *Ecology and Society* 14, p. 2.
- PATEL, M., KOK, K. & ROTHMAN, D.S. (2007) 'Participatory scenario construction in land use analysis: an insight into the experiences created by stakeholder involvement in the Northern Mediterranean', *Land Use Policy* 24, pp. 546–61.
- PETERSON, G.D., CUMMING, G.S. & CARPENTER, S.R. (2003) 'Scenario planning: a tool for conservation in an uncertain world', *Conservation Biology* 17, pp. 358–66.
- RASKIN, P., MONKS, F., RIBEIRO, T., VAN VUUREN, D. & ZUREK, M. (2005) 'Global scenarios in historical perspective', in Carpenter, S.R., Pingali, P.L., Bennett, E.M. & Zurek, M.B. (eds) *Ecosystems and human well-being*, Volume 2, *Scenarios findings of the Scenarios Working Group*, Island Press, Washington, DC, pp. 35–44.
- REED, M.S. (2008) 'Stakeholder participation for environmental management: a literature review', *Biological Conservation* 14, pp. 2417–31.
- ROBINS, L. (2008) 'Making capacity-building meaningful: a framework for strategic action', *Environmental Management* 42, pp. 833–46.
- ROBINS, L. & DOVERS, S. (2007) 'Community-based NRM boards of management: are they up to the task?', *Australasian Journal of Environmental Management* 14, pp. 111–22.
- ROBINSON, C.J., LANE, M.B. & TAYLOR, B.M. (2009) 'Conclusion: the changing and contested governance of Australia's environmental heritage', in Lane, M.B., Robinson, C.J. & Taylor, B.M. (eds) *Contested country: local and regional environmental management in Australia*, CSIRO, Canberra, pp. 243–5.
- ROBINSON, G.M. (1998) *Methods and techniques in human geography*, John Wiley, Chichester.
- ROTMANS, J., VAN ASSELT, M., ANASTASI, C., GREEUW, S., MELLORS, J., PETERS, S., ROTHMAN, D. & RIJKENS, N. (2000) 'Visions for a sustainable Europe', *Futures* 32, pp. 809–31.
- RYDIN, Y. (2003) *Conflict, consensus, and rationality in environmental planning: an institutional discourse approach*, Oxford University Press, Oxford.
- SALA, O.E., VAN VUUREN, D., PEREIRA, H.M., LODGE, D., ALDER, J., CUMMING, G., DOBSON, A., WOLTERS, V. & XENOPOULOS, M.A. (2005) 'Biodiversity across scenarios', in Carpenter, S.R., Pingali, P.L., Bennett, E.M. & Zurek, M.B. (eds) *Ecosystems and human well-being*, Volume 2, *Scenarios findings of the Scenarios Working Group*, Island Press, Washington, DC, pp. 375–408.
- SAYER, J. & CAMPBELL, B. (2004) *The science of sustainable development: local livelihoods and the global environment*, Cambridge University Press, Cambridge.
- SUSSKIND, L., MCKEARNAN, S. & THOMAS-LARMER, J. (1999) *The consensus building handbook: a guide to reaching agreement*, Sage, Thousand Oaks, CA.
- TACCONI, L. (2007) 'Decentralization, forests and livelihoods: theory and narrative', *Global Environmental Change* 17, pp. 338–48.
- WESTON, N. (2006) 'Local area planning approach to Mission Beach cassowary conservation stage 1: Framework' Cairns, Cooperative Research Centre for Tropical Rainforest Ecology and Management.
- WILLIAMS, K.J., HILL, R., PERT, P.L., HARDING, E.K. & O'MALLEY, T. (2008) 'Current, pre-clearing and 2025 scenarios of vegetation cover and cassowary habitat in Mission Beach and surrounds', in Taylor, R. & Long, S. (eds) *Proceedings of the 2008 Marine and Tropical Sciences Research Facility annual conference*, 28 April–1 May. Reef and Rainforest Research Centre Ltd., Cairns, pp. 130–49.
- WILMSEN, C., ELMENDORF, W., FISHER, L., ROSS, J., SARATHY, B. & WELLS, G. (2008) 'Participation, relationships and empowerment', in Wilmsen, C., Elmendorf, W., Fisher, L., Ross, J., Sarathy, B. & Wells, G. (eds) *Partnerships for empowerment: participatory research for community-based natural resource management*, Earthscan, London, pp. 259–84.
- WOLLENBERG, E., EDMUNDS, D. & BUCK, L. (2009) 'Interactive land-use planning in Indonesian rain-forest landscapes: reconnecting plans to practice', *Ecology and Society* 114, p. 35.